background

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1 Background

1.1 The basic pendulum

Let's start with the basic pendulum, a pivot which does not move and a single arm. We'll first find the equation of motion, a differential equation which when solved describes the motion of the object over time. Let's keep the shape of the pendulum general. Define the moment of inertia to be I, the mass to be m, the distance from the pivot to the center of mass to be L and its angle to be θ, where $\theta = 0$ means that the pendulum is hanging downward. There are two methods to do this:

1.1.1 Newtonian mechanics

About the pivot's axis, there is a torque

$$\tau = L \cdot (-mg \sin \theta),$$

and we know how torque relates to angular acceleration:

$$I\ddot{\theta} = \tau.$$

So out equation of motion is

$$\ddot{\theta} + \frac{mL}{I} g \sin \theta = 0.$$

Note that if you assume the arm consists of only a point mass, $mL/I = 1/L$.

1.1.2 Lagrangian mechanics

This method lets you not worry about forces, vectors, or free body diagrams. As long as you know the kinetic and potential energy, you can find the equation of motion.

The kinetic energy involved is the angular kinetic energy of the center of mass:

$$T = \frac{1}{2} I \dot{\theta}^2$$

and the potential energy comes from gravity, and we should express it with the same variable as in kinetic:

$$V = mgy = -mgL \cos \theta.$$

The Lagrangian is defined as $\mathcal{L} = T - V$. In our case,

$$\mathcal{L} = \frac{1}{2} I \dot{\theta}^2 + mgL \cos \theta.$$

To go from the Lagrangian to the equation of motion, we use the Euler-Lagrange equation for each degree of freedom q_i:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} = \frac{\partial \mathcal{L}}{\partial q_i}.$$

In our case, there's only one degree of freedom θ. So we get

$$\begin{aligned}
\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{\theta}} &= \frac{\partial \mathcal{L}}{\partial \theta} \\
\frac{d}{dt} I\dot{\theta} &= -mgL\sin\theta \\
I\ddot{\theta} &= -mgL\sin\theta \\
\ddot{\theta} + \frac{mL}{I} g\sin\theta &= 0
\end{aligned}$$

and we get the same answer!

1.1.3 Analyzing stability from the equation of motion

Even with such a simple equation of motion, we can't solve it analytically. However, we can still say something about its stability. First, it would be simpler if we only had to deal with first derivatives. Define ω to be the angular velocity, and note that $\dot{\omega} = \ddot{\theta}$. We have these two equations:

$$\begin{aligned}
\dot{\theta} &= \omega \\
\dot{\omega} &= -\frac{mL}{I} g\sin\theta
\end{aligned}$$

Let's define a vector $\mathrm{y} = \begin{pmatrix} \theta \\ \omega \end{pmatrix}$, and notice we can express our equation of motion with only a single derivative now:

$$\dot{\mathrm{y}} = \begin{pmatrix} \omega \\ -\frac{mL}{I} g\sin\theta \end{pmatrix}.$$